Name: …… Student ID: …….

The University of New South Wales

School of Electrical Engineering and Telecommunications

Final Examination

Term 3, 2021

ELEC3104: Digital Signal Processing

Candidates should attempt the questions as per the following instructions

1. This exam constitutes 35% of the total course assessments.
2. **Time allowed:** 2 hours (+ 10 min reading time + 20 minutes uploading answers)
3. You are required to answer only **4 questions**. However, if you answer more than 4 questions (which is NOT recommended), we will select the best 4 questions. **Please note that the maximum mark you can get is dependent on your choice of questions.**
4. **All** questions are presented in the SOLO Modified Taxonomy form:
    - The exam will be in the following layout:
        - Four questions under SOLO - Level 2 (4 x16 marks)
        - One question under SOLO - Level 3 (26 marks)
        - One question under SOLO - Level 4 (26 marks)
        - One question under SOLO - Level 5 (32 marks)
5. Print your name, student ID and question number on the front page of each **answer** page.
6. The questions are NOT of equal value. Part marks are as indicated.
7. Authorised examination materials: This is an open book exam.
8. Answers must be written in ink/digital ink or typed in MS Word. Upload the image/pdf of your answers on Moodle within the allocated time.
9. This paper contains 6 pages including this instruction page.

SOLO – Level 2: Question 1 [16 marks]

(a) An analogue signal x(t) = 3cos(100πt) + 2sin (250πt) is sampled 200 times per second.

(i) What are the frequencies in the resulting discrete-time signal x[n]? **[2 marks]**

(ii) Write down an equation for this discrete-time signal $x[n]$. **[2 marks]**

(iii) If $x[n]$ is passed through an ideal D/A converter, what is the reconstructed signal y(t)? **[2 marks]**

(b) Given a real-valued bandpass signal $x(t)$ with $X(f) = 0$ for $|f| < f_1$ and $|f| > f_2$ (see Figure 1), the Nyquist sampling theorem says that the minimum sampling frequency is $f_s = 2\,f_2$. However, in some cases, the signal may be sampled at a lower rate. Show that no aliasing will occur if $x(t)$ is sampled at a sampling frequency $f_s = 2\,(f_2 - f_1)$; where $f_2 = 3$ kHz and $f_1 = 2$ kHz. Sketch the magnitude response of the digital spectrum $X(\theta)$, where θ is the digital frequency ($-\pi \le \theta \le \pi$). **[6 marks]**

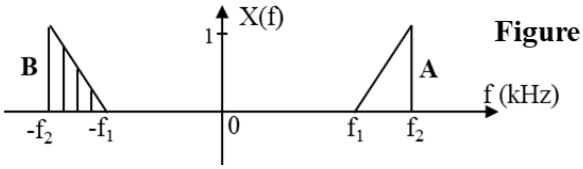


Figure 1

(c) (i) How many bits are required for the storage of a seismic signal that has a 60 second duration, if the sampling rate is $f_s = 20$ samples/sec and an 8-bit A/D converter is used? **[2 marks]**

(ii) What is the maximum frequency that can exist in the resulting digital seismic signal? **[2 marks]**

SOLO – Level 2: Question 2 [16 marks]

(a) A first-order digital low-pass filter is described by the system function

$$H(z) = \frac{k\,(1 + 0.5\,z^{-1})}{1 - 0.5\,z^{-1}}$$

(i) Determine k, so that the maximum value of $|H(\theta)|$ is equal to 1, where θ is the digital frequency. **[2 Marks]**

(ii) Show that the 3dB cut-off frequency, θ_c, of the filter $H(\theta)$ is given by $\theta_c = cos^{-1}(\frac{35}{44})$ **[6 Marks]**

(iii) Find the impulse response of the digital filter $H(z)$. **[4 marks]**

(b) Three systems with impulse responses $h_1[n] = \delta[n] - \delta[n-2]$, $h_2[n] = h_1[n]$ and $h_3[n] = u[n-1]$ are connected in cascade. Calculate the impulse response, $h_c[n]$, of the overall system?

[4 Marks]

SOLO – Level 2: Question 3 [16 marks]

(a) (i) A digital resonator is required to meet the following specifications:

1. Passband centred at $f_o = 50$ Hz **2**. Peak gain of unity at $f_o = 50$ Hz
3. 3-dB bandwidth of 6 Hz **4**. Sampling frequency (f_s) of 300 Hz,

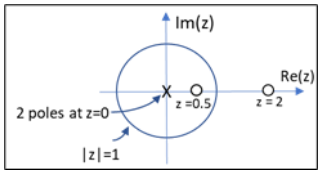
Using the pole-zero placement method, write the transfer function, $H(z)$, for the above digital resonator. **[7 marks]**

(ii) Draw an implementation of $H(z)$. **[3 marks]**

(b) A digital filter $H(z)$ has the pole-zero pattern as shown below. Its frequency response at $\theta = 0$ is $H(0) = 1$.

i) Write down the equation of the transfer function H(z). **[3 Marks]**

ii) Determine and sketch the phase response of the filter H(z). **[3 Marks]**



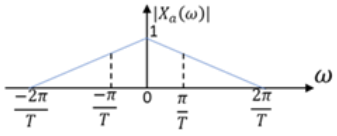
SOLO – Level 2: Question 4 (16 marks)

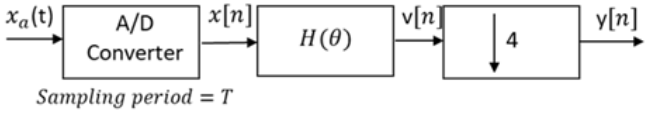
(a) (i) A signal has been sampled at 8kHz and for a particular application this signal is required to be used at an increased sampling rate of 12 kHz. Draw a system for digitally changing the sampling rate and state the bandwidth of the digital lowpass filter that is required to implement the system. [**4 Marks**]

(ii) For the system shown below $x_a(t)$ is the input analogue signal. It has the magnitude spectrum $|X_a(\omega)|$ as shown below. $H(\theta)$ is a digital lowpass filter and is given by

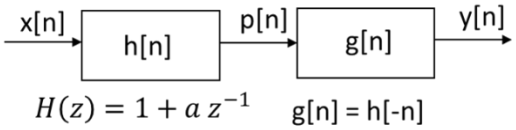
$$H(\theta) = \begin{cases} 4 & |\theta| \le 1 \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases}$$

Sketch $|X(\theta)|, |V(\theta)|$ and $|Y(\theta)|$ against θ; where θ is the digital frequency and ω is the analogue frequency. [**8 Marks**]





(b) Show that if $x[n]$ is a real value signal, the structure shown below has a zero-phase response, where a $(0 < a < 1)$ is the filter coefficient of the filter $H(z)$. [**4 Marks**]

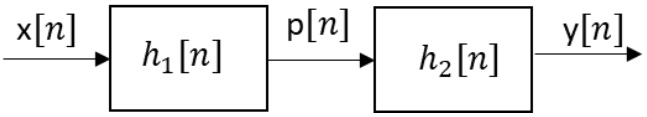


End of SOLO - Level 2 Examination Questions

SOLO – Level 3: Question [26 marks]

(a) Consider the following interconnection of a linear time invariant (LTI) systems, where $h_1[n] = \delta[n] - \delta[n-1]$. The DTFT of $h_2[n]$ is given by $H_2(\theta)$:

$$H_2(\theta) = \begin{cases} 1 & |\theta| \leq \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \leq \pi \end{cases}$$



(i) Show that $h_2[n] = \frac{\sin\left(\frac{n\pi}{2}\right)}{n\pi}$ **[4 marks]**

(ii) Show that the impulse response $h[n]$ of the overall system is given by

$$\text{h[n]} = \frac{\sin\left(\frac{n\pi}{2}\right)}{n\pi} - \frac{\sin\left[\frac{(n-1)\pi}{2}\right]}{(n-1)\pi}$$

[6 marks]

(iii) Show that the frequency response $H(\theta)$ of the overall system is given by

$$H(\theta) = \begin{cases} 1 - e^{-j\theta} & |\theta| \leq \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \leq \pi \end{cases}$$

[4 marks]

(b) (i) Use the bilinear transform to design an IIR digital lowpass filter based on the third order Butterworth prototype normalised function:

$$\text{G(s)} = \frac{1}{(\text{s}^2+\text{s}+1)(\text{s}+1)}$$

The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3 dB cut-off frequency at 2 kHz. Show that G(z) reduces to: **[8 marks]**

$$\text{G}(z) = \frac{(\text{z}+1)^3}{6\text{z}\,(\text{z}^2 + \frac{1}{3})}$$

(ii) Draw an implementation of this filter G(z) as a cascade structure. **[4 marks]**

End of SOLO - Level 3 Examination Question.

SOLO – Level 4: Question 1 [26 marks]

(a) In applications involving the digital modulation of sinusoidal carrier signals, it is necessary to generate sine and cosine signals simultaneously from an oscillator. Consider the system

$$y[n] = a\,y[n-1] + x[n]$$

where $a = e^{j\theta_o}$ and the frequency of oscillation is θ_o. Let the input $x[n]$ be real and the output $y[n]$ is complex. Thus

$$y[n] = y_R[n] + j\,y_I[n]$$

(i) Determine the equations describing a system with one input $x[n]$ and two outputs $y_R[n]$ and $y_I[n]$ only. **[6 Marks]**

(ii) Draw a structure for the realisation of this oscillator. **[4 Marks]**

(iii) Show that if $x[n] = \delta[n]$ then

$$y_R[n] = cos[n\theta_0].u[n] \text{ and } y_I[n] = sin[n\theta_0].u[n]$$

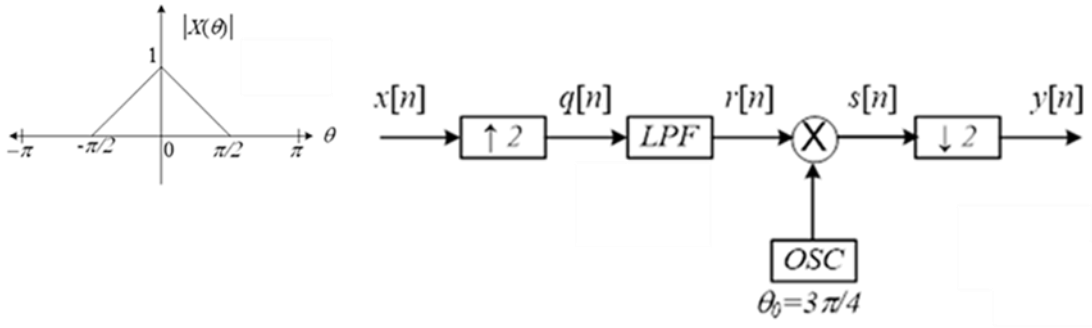
[4 Marks]

(b) A modulator scheme is shown below where the digital input $x[n]$ is sampled at a sampling frequency of 8kHz and the magnitude spectrum |X(θ)| of $x[n]$ is also given below. The carrier frequency of the cosine oscillator is $\theta_o = 3\pi/4$.

The ideal lowpass filter $H_d(\theta)$, denoted as LPF in the modulation scheme in the frequency domain is given by

$$H_d(\theta) = \begin{cases} 4 & 0 \le |\theta| \le \pi/4 \\ 0 & otherwise \end{cases}$$

If |X(θ)| is the magnitude spectrum of $x[n]$, sketch the magnitude spectra of |Q(θ)|, |R(θ)| , |S(θ)| and |Y(θ)| in separate diagrams. **[12 Marks]**



End of SOLO - Level 4 Examination Question.

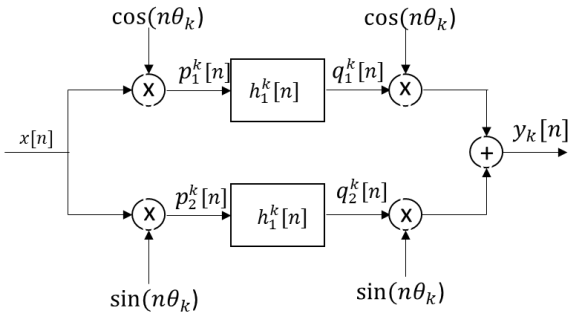
SOLO – Level 5: Question [32 marks]

(a) (i) Show that the impulse response of the system for the kth channel (shown below), is

$$h_k[n] = h_1^k[n]\ cos(n\theta_k)$$ **[6 marks]**

(ii) Show that the frequency response $H_k(\theta)$ of the system is given by

$$H_k(\theta) = \frac{1}{2}H_1^k(\theta - \theta_k) + \frac{1}{2}H_1^k(\theta + \theta_k)$$ **[6 marks]**

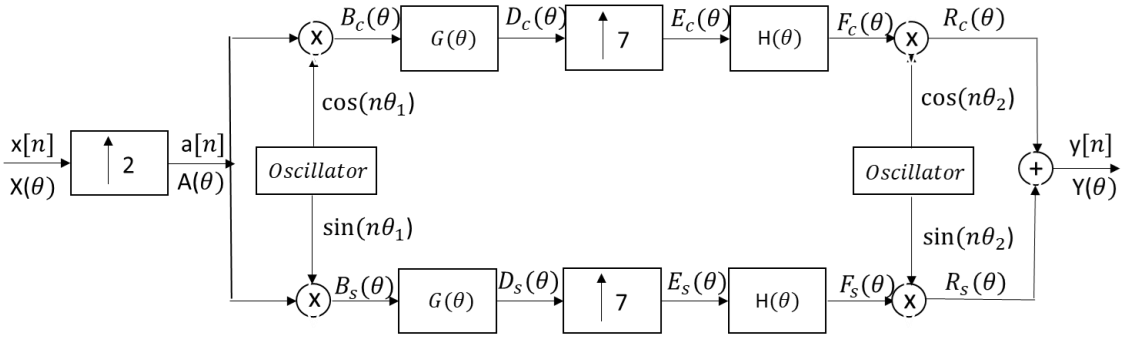


(b) The diagram below shows a digital modulator. The sampling period T of the input signal $x[n]$ is 125 μs. The first oscillator generates a carrier of frequency θ_1. The second oscillator generates a carrier of frequency θ_2: $\theta_1 = \omega_1\frac{T}{2}$ and $\theta_2 = (\omega_c + \omega_1)\frac{T}{2}$;

where $\omega_1 = 2\pi\,.\,2\,.\,10^3\ rad/sec$ and $\omega_c = 2\pi\,.\,4\,.\,10^3\ rad/sec$.

The low-pass filters G(θ) and H(θ) have the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases} \qquad H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{28} \\ 0 & \frac{\pi}{28} < |\theta| \le \pi \end{cases}$$



If $X(\theta)$ = 1+cos(θ), by writing appropriate equations show that

(i) $$D_c(\theta) = \begin{cases} 1 & -\pi/4 \le \theta \le \pi/4 \\ 0 & elsewhere \end{cases} \qquad D_s(\theta) = \begin{cases} \frac{1}{j}\sin(2\theta) & -\pi/4 \le \theta \le \pi/4 \\ 0 & elsewhere \end{cases}$$ **[6 marks]**

(ii) Sketch the magnitude spectrum of each $F_c(\theta)$, $R_c(\theta)$, $D_s(\theta)$, $F_s(\theta)$, $R_s(\theta)$ and Y(θ) against θ in the fundamental frequency region $-\pi \le \theta \le \pi$. **[14 marks]**

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End of SOLO - Level 5 Examination Question.